

CHAPTER

2

Some facts about prosperity and growth

An *Inquiry into the Nature and Causes of the Wealth of Nations*. This was the title Adam Smith gave his famous book published in 1776, a book which greatly advanced economics as a scientific discipline. Already in the title Adam Smith stated what he considered to be the most important issue in economics: what is it that makes a nation prosperous, to the benefit of its citizens? Smith was clear from the very beginning about how prosperity should be measured:

The annual labour of every nation is the fund which originally supplies it with all the necessaries and conveniences of life which it annually consumes, and which consists always either in the immediate produce of that labour, or in what is purchased with that produce from other nations.

According therefore as this produce, or what is purchased with it, bears a greater or smaller proportion to the number of those who are to consume it, the nation will be better or worse supplied with all the necessaries and conveniences for which it has occasion.

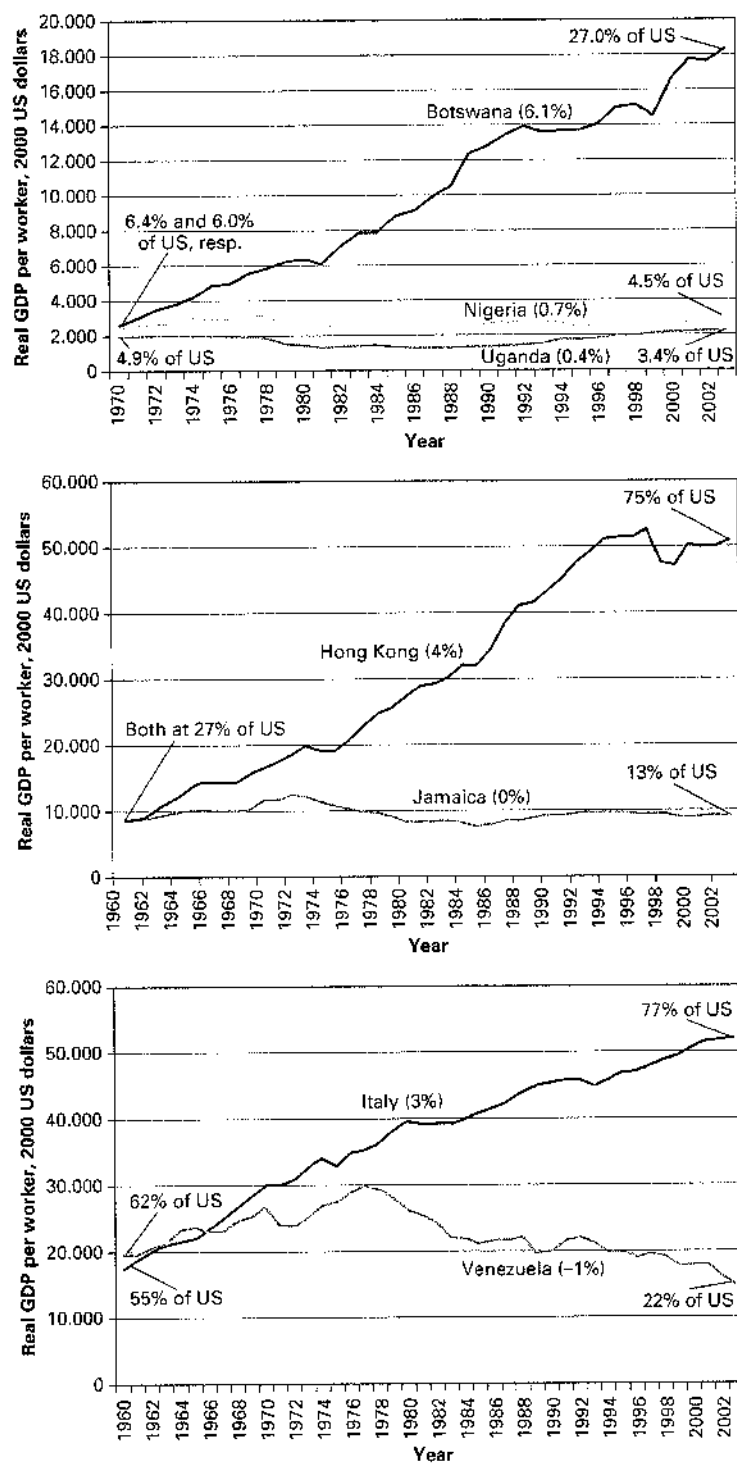
(Opening remarks of 'Introduction and Plan of the Work' of *The Wealth of Nations*, 1776).

In modern economic language, Smith says that the average level of prosperity in a country can be measured by the country's GDP or income per person. It is not necessarily all of the annual income that is consumed during the year, but what is not consumed is saved, and becomes either investment or an export surplus. In both uses it adds to the national wealth and thereby becomes a source of future consumption. Consumption is thus always rooted in production and income, and in so far as consumption is a good proxy for the economic well-being of people, annual GDP or income per person is a relevant measure of prosperity. Clearly, not only the average income per person but also the *distribution* of income across persons is of interest. However, for a given degree of income inequality, an increase in average GDP per person is to the benefit of everybody.

According to Adam Smith's view, it is the annual *level* of GDP per person that is of importance. Nevertheless, in Book One we will be much concerned with economic growth, i.e. the annual *increase* in GDP per person. We are interested in growth, not as an end in itself, but because *the way a country can reach a higher level of income is through a process of growth*.

Figure 2.1 gives some clear illustrations of the importance of gradual growth for reaching higher, or falling to lower, levels of income per person.¹ Around 1970, three African nations,

1. The figure shows GDP per worker (GDP per person in the labour force), not GDP per capita (GDP per person in the population) for reasons to be explained.

FIGURE 2.1 The importance of growth in GDP per worker for the level of GDP per worker

Source: Penn World Table 6.2.

Botswana, Nigeria and Uganda, were all at about the same level of GDP per worker, and this level was very low by international standards. Over the subsequent 33 years, GDP per worker in Botswana grew at an impressive average rate of about 6 per cent per year, while the growth rates of Nigeria and Uganda were 0.7 per cent and 0.4 per cent, respectively. This implied that in 2003 the *level* of GDP per worker in Botswana was 6 times greater than in Nigeria and 8 times greater than in Uganda. By 2003 Botswana had not become extremely rich, but rather than having, like Nigeria and Uganda, a GDP per worker at or below 4.5 per cent of the US level, Botswana had reached a level of 27 per cent of US income per capita. The differences in the conditions of life caused by these different growth experiences are not difficult to imagine.

Both Jamaica and Hong Kong had a GDP per worker of around 27 per cent of that in the USA in 1960. From that time until 2003, Jamaica grew at an average rate of 0.1 per cent a year, while Hong Kong grew at an annual rate of 4.2 per cent. Today Hong Kong is one of the world's rich areas, with a GDP per worker of about 75 per cent of the US level in 2003, while Jamaica is relatively poor at around 13 per cent of the US level.

In 1960 Venezuela was one of the world's rich countries with a GDP per worker of 62 per cent of the US level, while Italy was considerably behind at 55 per cent of the US level. However, until 2003, average annual growth in Italy was 2.6 per cent, but in Venezuela it was negative. In 2003 it is Italy, not Venezuela, that is one of the world's rich countries, with a GDP per worker at 77 per cent of the US level. Venezuela has fallen to 22 per cent of the US level.

What allows a country to escape from poverty, or to achieve prosperity, is a process of growth over a succession of years. Therefore one of the most important questions in economics, probably *the* most important one, is: what creates growth? Growth theory and growth empirics are fascinating subjects in economics. We start in this chapter by stating and discussing some empirical regularities concerning prosperity and growth. These 'stylized facts' constitute important knowledge in themselves. They will also be used later on as yardsticks: the growth theory to be presented will be held up against the empirical facts.

2.1 Measuring the wealth of a nation

Agreeing with Adam Smith that the ratio of the annual 'produce' to 'the number of those who are to consume it' is a relevant measure of the average standard of living in a nation, we will use GDP per person as a proxy for living standards. You are assumed to know about statistics such as GDP, total consumption, total investment, etc., from an earlier course in macroeconomics. We will therefore discuss only a few aspects of measuring economic welfare by GDP per person.

Suppose we would like to compare the standard of living between a highly developed country such as the USA, and a somewhat less developed country with relatively fewer transactions in the official market economy and relatively more self-sufficiency. The GDP of each country is the value of the 'official' marketed production.

First, a comparison between the two countries requires that their GDPs be measured in the same currency, say US dollars. For this purpose we need an exchange rate that converts amounts denominated in the currency of the less developed country, the peso say, into amounts in dollars. It may be misleading to use the official prevailing exchange rate between the peso and the dollar since prevailing exchange rates are volatile. A 10 per cent increase in the peso relative to dollars from one month to the next will typically not reflect that the less developed country has become 10 per cent richer compared to the USA in one month. It will rather be a reflection of some change in expectations. Current exchange rates are not appropriate for converting GDPs if one is interested in standards of living. Instead one should use a rate of conversion that reflects purchasing power. The total cost of a relevant bundle of goods representing the 'necessaries and conveniences' of life should be computed in the two countries, and the purchasing-power-adjusted exchange rate could then be defined as the

relation between the two total costs. If the bundle has a total cost of 1000 dollars in the USA, and a cost of 100,000 pesos in the less developed country, the purchasing-power-adjusted exchange rate is one dollar for 100 pesos. One should use this computed exchange rate to convert the GDP of the less developed country into a purchasing-power-adjusted GDP in US dollars.

A second issue is whether GDP per person should mean GDP per capita, where the GDP is divided by the size of the total population, or whether it should mean GDP per worker, where the GDP is only divided by the size of the labour force, which is the population size times the labour force participation rate. The less developed country will typically have a larger informal sector and relatively more people living from non-marketed home production. If one divides the official GDP of the country, the value of the marketed production, by the size of the entire population, one will probably underestimate the prosperity of the country in comparison with the highly developed country. Dividing the official GDP by the size of the official labour force can be a way of correcting for this bias, since the official labour force does not include those working only in the informal sector. (If the productivity in the formal and informal sectors are identical, the correction should be perfect.) Generally, using GDP per worker as our measure of the standard of living makes at least some correction for cross-country differences in the degree of participation in the formal economy.

Table 2.1 shows that it may indeed make a difference whether one considers GDP per capita or per worker. For instance, measured by GDP per capita, Egypt and Pakistan are only 13 per cent and 7 per cent as rich as the USA in the year 2000, respectively. Measured by GDP per worker, they are still not rich compared to the USA, but they are considerably less poor, now at 18 per cent and 10 per cent of the US level, respectively. This reflects the low participation rates in Egypt and Pakistan.

According to GDP per capita, Ireland was 73 per cent as rich as the USA in 2000, but measuring by GDP per worker, Ireland was 88 per cent as rich. The difference is a reflection of the much higher participation rate in the USA. Also note that according to GDP per capita, Denmark is richer than the Netherlands, but according to GDP per worker it is a bit poorer, reflecting the higher participation rate in Denmark. For the examples of Ireland vs. the USA, and Denmark vs. the Netherlands, the differences in participation rates are probably not caused by differences in the relative sizes of the unofficial sectors, but rather by 'true' differences in labour force participation, e.g. by women. Nevertheless, it can still be argued

TABLE 2.1 GDP per capita and per worker (2000)

Country	Absolute, 2000 US dollars		Implicit participation rate ¹	Relative to US	
	GDP per capita	GDP per worker		GDP per capita	GDP per worker
USA	34,365	67,079	0.51	1.00	1.00
Denmark	27,827	50,448	0.55	0.81	0.75
Japan	23,971	44,563	0.54	0.70	0.66
Netherlands	26,293	56,691	0.46	0.77	0.85
Belgium	24,662	59,874	0.41	0.72	0.89
Sweden	25,232	46,544	0.54	0.73	0.69
UK	24,666	49,225	0.50	0.72	0.73
Ireland	24,948	59,103	0.42	0.73	0.88
Egypt	4,536	11,940	0.38	0.13	0.18
Pakistan	2,477	6,719	0.37	0.07	0.10

¹ Computed as GDP per capita divided by GDP per worker

Source: Penn World Table 6.2

that GDP per worker is a more appropriate measure of economic standards of living than GDP per capita, because home production and leisure should also count.

GDP per worker is closer to a *productivity* measure than GDP per capita. In the growth models presented in the subsequent chapters, average labour productivity is a key variable, and productivity has a closer correspondence to GDP per worker than to GDP per capita. Since we want a close correspondence between our empirical and theoretical variables, this is another reason for studying GDP per worker.²

The statistics used for international comparison in this book are of the purchasing-power-adjusted type, and we will mainly use GDP per worker as a proxy for the standard of living and for labour productivity. Data will, to a large extent, come from a database called the Penn World Table (PWT), which is an attempt to create internationally comparable statistics for the countries of the world for a considerable period (for many developed countries 1950–2003, and for most other countries 1960–2003).³ The PWT does not cover all countries in the world. Among the exceptions are nations that were formerly part of the Soviet Union and other formerly planned economies of Eastern Europe.

2.2 The rich and the poor, the growing and the declining

We have already given some illustrations of the enormous differences between rich and poor countries, and of the fact that some countries have managed to develop from poor to not so poor through a good growth performance. We now go into more detail about the issues of poverty and prosperity, and about moving in and out of these categories.

The world income distribution

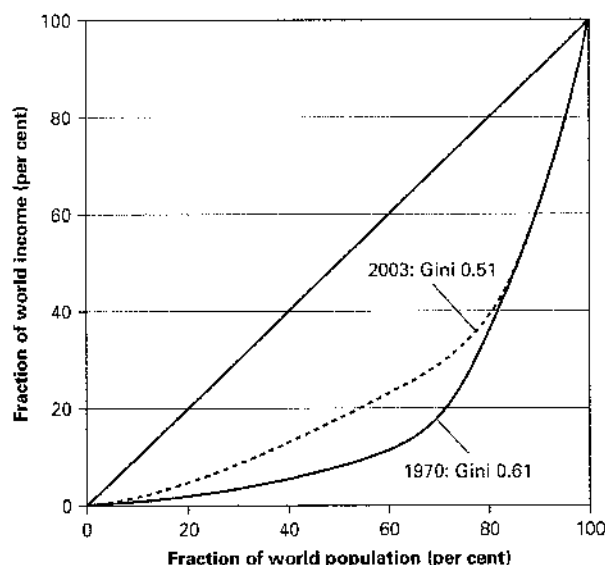
Some countries are poor and some are rich, but is there any sign that the world income distribution has become more equal? We can investigate this by means of so-called Lorenz curves. Figure 2.2 shows two Lorenz curves for the world, one for 1970 and one for 2003. A point (x, y) on a Lorenz curve indicates that the fraction x of the people in the world with the smallest incomes earns the fraction y of all the income in the world. For instance, the Lorenz curve for 1970 shows that in that year, the 60 per cent of the world population that lived in the poorest countries earned a bit more than 10 per cent of the world's income. In 2003 the corresponding figure was considerably above 20 per cent.

The curves are based on average GDP per worker and assume that *all* people living in a specific country, not just all people in the labour force, have an income equal to the average per worker GDP recorded in the PWT, in the country and for the year in question. We thus attribute to each individual in a country an artificial income equal to the recorded average GDP per worker. The Lorenz curves have been constructed from such personal incomes and population sizes (you will be asked to do a similar construction yourself in Exercise 2). The procedure may seem a bit odd, but in this way we can base the Lorenz curves on data for average GDP per worker.

Absolute equality corresponds to a Lorenz curve identical to the 45° line, or diagonal. The further the Lorenz curve is below the diagonal, the more unequal is the income distribution. Therefore the area between the diagonal and the Lorenz curve, as a fraction of the entire area below the diagonal, is an aggregate measure of inequality. This measure, called the Gini coefficient, is a number between zero and one, and the closer it is to one the more inequality

2. By this argument one should perhaps go all the way to GDP per work hour to allow not only for differences in participation rates, but also for differences in annual working time and unemployment rates. Lack of good internationally comparable statistics for annual work hours prevents this.

3. The PWT can be found on the internet at <http://pwt.econ.upenn.edu/>. At the time this second edition was prepared (2008), the most current version was PWT 6.2, from which most growth data in this book are taken. An extract from the PWT 6.2, and from other relevant databases, can be found in Table A at the end of Book One.

FIGURE 2.2 Lorenz curves for the world, 1970 and 2003, 137 countries

Source: Penn World Table 6.2.

there is. The Gini coefficients for 1970 and 2003 are shown in Fig. 2.2, and they are 0.61 and 0.51, respectively. Because of the particular construction of the Lorenz curves based on GDP per worker, these Gini coefficients are not comparable to traditional coefficients computed for the income distribution of a specific country. However, they should be comparable to each other.

Investigating the Lorenz curves and the Gini coefficients reveals that there were enormous income differences in the world both in 1970 and in 2003, but between these two years the world income distribution *does* seem to have become more equal.

Our investigation of the world income distribution is a bit crude. Most of the improvement indicated by Fig. 2.2 comes from the relatively good growth performances of some heavily populated countries, most importantly China and India (as illustrated in Table 2.3 below). This does not necessarily mean that Fig. 2.2 gives a distorted impression, but in the construction we have completely neglected the influence of inequality *within* countries by attributing the same income to all persons in each country. If, for instance, the personal income distribution in China has become more unequal over the period considered, Fig. 2.2 will exaggerate the degree to which the *personal* world income distribution has become more equal. It is interesting to note that more thorough studies of the evolution in the world income distribution, which account for inequality within countries, reach conclusions similar to the one appearing from our more crude analysis.⁴

4. One study is by economist Xavier Sala-i-Martin: 'The World Distribution of Income: Falling Poverty and . . . Convergence, Period', *Quarterly Journal of Economics*, 121, 2006, pp. 351–397. Sala-i-Martin emphasizes strongly the overall improvement in the world income distribution from 1970 to 2000, achieved mainly over the two decades from 1980 to 2000, but he also expresses a serious concern about 'the dismal growth performance of the African continent', writing: '... where poverty was mostly an Asian phenomenon 30 years ago ... poverty is, today, an essentially African problem'.

STYLIZED FACT 1

Some countries are rich and some are poor, the differences are enormous, and it has pretty much stayed like that in relative terms for a long time. However, there is some tendency towards a more equal world income distribution over the past three to four decades, but not much at the very bottom.

Two features should be noted. First, the Lorenz curve and the associated Gini coefficient illustrate the income distribution in *relative* terms. If the Lorenz curve had been completely unchanged from 1970 to 2003 it would mean that the percentage change in income per capita in the world's poorest country would have been the same as in the richest country. In absolute terms the poorest country would be much less poor by 2003 than in 1970. Indeed, the world average income per worker (computed with population sizes as weights) increased from 1970 to 2003 by an average annual rate of 1.4 per cent, corresponding more or less to 60 per cent over the full period. The fact that the Lorenz curve is relatively unchanged at the very bottom means that the poorest countries have also enjoyed most of this increase. On the other hand, if the world income distribution is unchanged in relative terms, and incomes generally increase, then the *absolute* differences between the poorest and the richest increase.

Second, it is not exactly the same countries that are the poorest (or the richest) in 1970 and 2003. The fact that the 10 per cent of people living in the poorest countries earned only around 1.5 per cent of world income in 1970 as well as in 2003 does not mean that it is impossible to escape from poverty. There were some countries that did move out of the group of the poorest countries by having high growth (compared to the world). However, the other countries that took their place among the poorest had sufficiently bad growth experiences to ensure that the group of poorest countries did not increase their share of the world's income.

Prosperity and poverty, growth and decline

The movement in and out of the groups of the poorest and of the most prosperous countries is illustrated in Table 2.2. The table states some statistics for the countries that were among the 15 poorest in terms of GDP per worker in 1965 and in 2003, and for the 15 richest countries in these years. The average annual growth rates between 1965 and 2003 reported in Table 2.2 have been calculated as $(\ln y_{03} - \ln y_{65})/38$, where y_t is GDP per worker in year t : if the relative change in a variable is not too large, it is approximately equal to the variable's absolute change in (natural) logs.⁵

Table 2.2 illustrates again the enormous differences between the rich and the poor, but also a fair amount of income mobility. Some countries are among the 15 poorest both in 1965 and 2003, but other countries have left the group of the poorest. For instance, in 1965 China was among the poorest with a level of GDP per worker at 2.4 per cent of that in the USA. By 2003 China had advanced to 12.2 per cent of the US level. This was achieved by an average annual growth rate in GDP per worker of 6.1 per cent, bringing China safely out of the group of the 15 poorest nations in 2003, which all had a GDP per worker of less than 3.5 per cent of the US level.

Other countries that moved out of the bottom 15 are, for example, Lesotho and Ghana. Over the same 38 years countries such as Niger and Zambia moved the other way, into the group of the poorest, by having bad growth performances. Their GDP per worker relative to the US fell from around 7 per cent to 3.3 per cent or less.

5. The rate of change $(z - x)/x = z/x - 1$ is approximately equal to $\ln z - \ln x = \ln z/x$, since, as long as a number u is not too far from one, then $\ln u \approx u - 1$. The latter follows because $\ln 1 = 0$, and $d \ln u / du = 1/u$, so measured at $u = 1$, the slope of the log function is one. Hence, locally around $(1, 0)$, the function $\ln u$ goes like the function $u - 1$. If the relative change we are interested in is a growth rate in, e.g., GDP per worker, $(y_t - y_{t-1})/y_{t-1}$, then typically it is not too far from zero, so y_t/y_{t-1} is close to one, and then $(y_t - y_{t-1})/y_{t-1} \approx \ln y_t - \ln y_{t-1}$ is a good approximation.

TABLE 2.2 The world's prosperity: top 15 and bottom 15, 1965 and 2003

	Real GDP per worker relative to USA		Average annual growth rate 1965–2003		Real GDP per worker relative to USA		Average annual growth rate 1965–2003
	Per cent	Per cent			Per cent	Per cent	
Poorest in 1965	1965	2003	Per cent	Poorest in 2003	1965	2003	Per cent
Ghana	1.9	4.2	3.8	Guinea-Bissau	3.4	1.9	0.1
China	2.4	12.2	6.1	Burundi	3.1	2.1	0.6
Malawi	2.5	2.4	1.4	Ethiopia	2.5	2.3	1.3
Ethiopia	2.5	2.3	1.3	Madagascar	6.5	2.3	–1.1
Tanzania	2.8	2.6	1.4	Malawi	2.5	2.4	1.4
Burkina Faso	2.9	3.4	2.0	Tanzania	2.8	2.6	1.4
Burundi	3.1	2.1	0.6	Gambia	4.3	2.7	0.3
Mali	3.1	3.6	2.0	Niger	6.8	2.7	–0.9
Guinea-Bissau	3.4	1.9	0.1	Togo	6.2	2.7	–0.6
Nepal	4.1	4.5	1.9	Chad	6.3	2.9	–0.5
Rwanda	4.2	3.5	1.1	Zambia	6.9	3.3	–0.4
Lesotho	4.2	7.0	2.9	Burkina Faso	2.9	3.4	2.0
Mozambique	4.3	4.1	1.5	Uganda	4.7	3.4	0.7
Gambia	4.3	2.7	0.3	Kenya	6.8	3.4	–0.2
Uganda	4.7	3.4	0.7	Rwanda	4.2	3.5	1.1
Richest in 1965	1965	2003		Richest in 2003	1965	2003	
Switzerland	103.0	79.9	0.9	United States	100.0	100.0	1.6
United States	100.0	100.0	1.6	Ireland	44.0	97.1	3.7
New Zealand	99.1	85.3	0.5	Norway	76.4	96.8	2.2
Netherlands	91.5	83.7	1.4	Belgium	71.4	90.7	2.2
Canada	87.5	76.3	1.2	Austria	60.2	88.1	2.6
Denmark	81.9	75.9	1.4	France	67.0	83.9	2.2
Sweden	81.2	70.9	1.2	Netherlands	91.5	83.7	1.4
Australia	78.7	80.5	1.7	Australia	78.7	80.5	1.7
Norway	76.4	96.8	2.2	Singapore	35.7	80.0	3.8
Belgium	71.4	90.7	2.2	Switzerland	103.0	79.9	0.9
Iceland	71.2	68.1	1.5	Italy	59.1	76.8	2.3
United Kingdom	67.6	78.5	1.9	United Kingdom	67.6	76.5	1.9
France	67.0	83.9	2.2	Canada	87.5	76.3	1.2
Israel	66.5	69.9	1.7	Denmark	81.9	75.9	1.4
Venezuela	63.5	21.9	–1.2	Hong Kong	37.1	75.2	3.5

Note: Countries are chosen from a sample of 97 countries, for which data on real GDP per worker in 1965 and 2003 were available. A lower limit of 1 million citizens was used.

Source: Penn World Table 6.2.

At the rich end there was also a substantial amount of mobility. Most impressively, Hong Kong, Singapore and Ireland, starting at levels of GDP per worker relative to the USA of 37, 36 and 44 per cent, respectively, moved into the top 15, and ended at levels 75, 80 and 97 per cent, respectively. Most spectacular drop-outs of the top 15 are Venezuela and New Zealand.

TABLE 2.3 World growth: bottom 15 and top 15 1965–2003

	Average annual growth rate of GDP per worker 1965–2003	Real GDP per worker relative to USA,	
		Per cent	Per cent
15 slowest growing	Per cent	1965	2003
Nicaragua	–1.9	48.9	12.7
Jordan	–1.3	50.8	16.8
Venezuela	–1.2	63.5	21.9
Madagascar	–1.1	6.5	2.3
Niger	–0.9	6.8	2.7
Togo	–0.6	6.2	2.7
Chad	–0.5	6.3	2.9
Zambia	–0.4	6.9	3.3
Peru	–0.3	33.9	16.3
Jamaica	–0.3	27.3	13.3
Kenya	–0.2	6.8	3.4
Senegal	–0.2	9.1	4.6
Bolivia	0.0	19.2	10.7
Guinea-Bissau	0.1	3.4	1.9
El Salvador	0.1	28.8	16.1
15 fastest growing			
China	6.1	2.4	12.2
Korea, Republic of	5.2	13.4	49.8
Thailand	4.2	6.9	18.0
Malaysia	4.0	16.9	41.2
Ghana	3.8	1.9	4.2
Singapore	3.8	35.7	80.0
Ireland	3.7	44.0	97.1
Romania	3.6	8.7	18.5
Hong Kong	3.5	37.1	75.2
Japan	3.2	37.0	66.4
Sri Lanka	3.1	7.7	13.7
Indonesia	3.1	7.1	12.2
India	2.9	6.0	9.9
Lesotho	2.9	4.2	7.0
Austria	2.6	60.2	88.1

Note: Sample of 97 countries as for Table 2.2. Botswana is missing for lack of 1965 to 1970 data, while Taiwan is missing for lack of 1993–2003 data. They would otherwise have been in the top of the top 15.

Source: Penn World Table 6.2.

Table 2.2 reveals another interesting, and perhaps surprising, fact: even among the world's 15 richest countries in 2003, there are substantial differences in income per worker, with the lowest levels of GDP per worker at around 75 per cent of the US level.

Since a country can move from poor to rich through a process of growth, and vice versa, it is of interest to look directly at growth performances. Table 2.3 shows the bottom 15 and the top 15 for growth in GDP per worker between 1965 and 2003. It is not surprising to find

the so-called 'tiger economies' of East Asia in the top 15: South Korea, Thailand, Malaysia, Singapore, Hong Kong, and also Japan. It is perhaps more surprising to find the African countries Ghana and Lesotho in the top 15 (together with Botswana, see the note). The appearances of China and India should no longer be surprising these days.

The countries at the top of the top 15 are often called 'growth miracles'. Likewise, the countries in the bottom 15 can be called 'growth disasters'. Note that for many of these countries, average growth rates between 1965 and 2003 were actually negative.

The countries of the world that are neither in the growth top 15 nor the bottom 15 have had annual growth rates between the 2.6 per cent of the slowest-growing countries in the top 15 and the 0.1 per cent of the fastest-growing countries in the bottom 15; see for instance Table A at the end of Book One. Our insights can be summarized as follows:

STYLIZED FACT 2

Growth rates vary substantially between countries, and by the process of growing or declining quickly, a country can move from being relatively poor to being relatively rich, or from being relatively rich to being relatively poor.

Hidden in the data is another interesting fact. Consider the tiger economies, South Korea, Thailand, Malaysia, and Singapore (Taiwan could be included, but does not appear in Table 2.3 because of lack of data), which all grew very fast between 1965 and 2003, with average annual growth rates of GDP per worker in the range from 4 per cent and up. Up to around 1960 these countries grew much more slowly. In fact, in the 1960s some economists worried about the prospects for countries in South Asia.⁶ All the growth miracles of South-East Asia experienced breaks in their growth rates from lower to higher values somewhere around the early 1960s. Similar growth breaks have occurred for several other countries in the growth top 15.

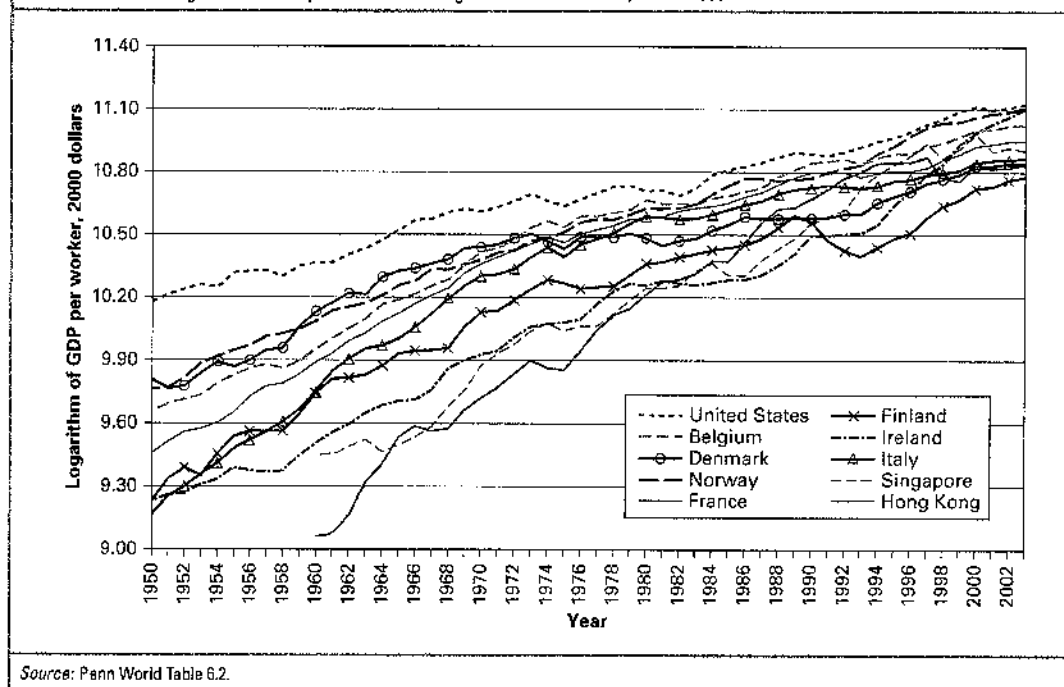
STYLIZED FACT 3

Growth can break in a country, turning from a high rate to a low one or vice versa.

You may perhaps think that this is a trivial fact: there is nothing inevitable about growth. However, it illustrates that the fight against poverty is not a hopeless one. In particular, one cannot point to a continent or part of the world in which the break in growth from low to high should be impossible. We stress this because sometimes in the public debate it is taken for granted that countries in some parts of the world, sub-Saharan Africa or the Middle East for instance, will never be able to grow out of poverty. The data simply do not support such an idea. There are no growth-preventing areas in the world, but there may be growth-preventing policies or circumstances (civil war, for instance).

Stylized fact 3 raises again the perhaps most important question in economics: *how* can a country create a positive growth break? What kinds of policies can do that? Growth theory and empirics, as presented in the coming chapters, can offer some possible answers.

6. In 1968 the Nobel Prize Laureate Gunnar Myrdal published the book (in three volumes, covering more than 2000 pages), *Asian Drama: An Inquiry into the Poverty of Nations*. Allen Lane, Penguin Press, London. The book was, as the title suggests, much concerned with why countries in South Asia, and in particular in South-East Asia, were not able to escape from poverty by a rapid growth process. At the time of publication several of these countries, but certainly not all, had started to grow fast.

FIGURE 2.3 Convergence of GDP per worker among selected countries, 1950–2003

Source: Penn World Table 6.2.

2.3 Convergence

An interesting idea in economics would, if it were true, imply that poverty should disappear by itself.

Absolute convergence

Figure 2.3 plots the log of annual GDP per worker for the period 1950 to 2003 for a selection of countries. A glance at the figure suggests that all the countries tend to approach one and the same 'growth path'. Since $\ln y_t - \ln y_{t-1}$ is (approximately) the rate of growth in y_t , in a figure showing $\ln y_t$ against time t , constant growth corresponds to a straight line, and the slope of the line is the growth rate. Figure 2.3 suggests a tendency for countries to approach one and the same upward-sloping straight line, perhaps more or less a line corresponding to the USA.

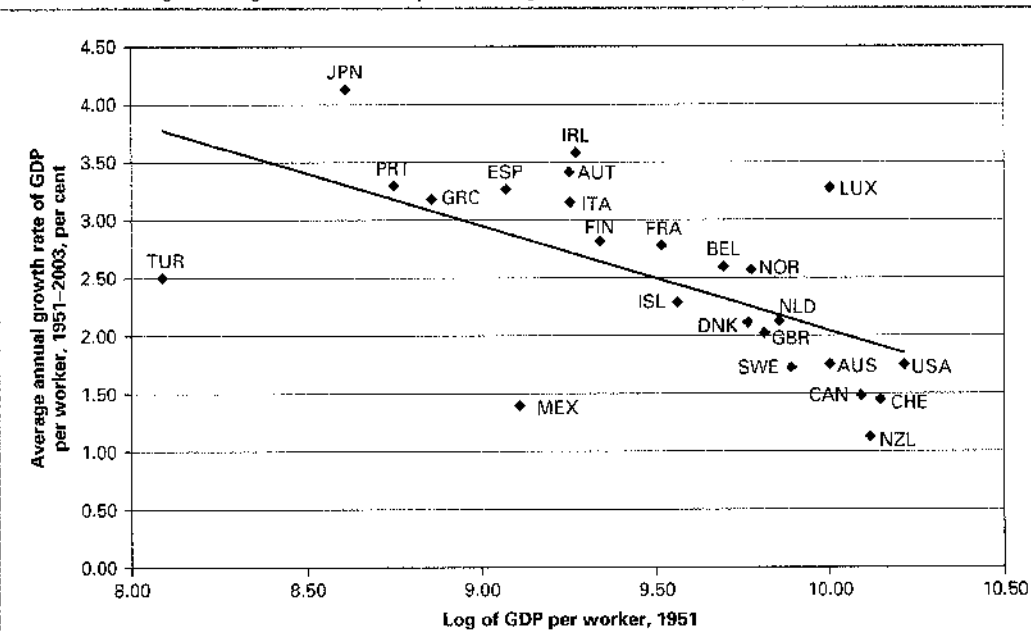
Figures like Fig. 2.3 (drawn for even longer periods, up to 100 years and more) were presented by William J. Baumol in a famous article from 1986, suggesting some of the 'laws' discussed below.⁷ Such figures open up the fascinating possibility that the differences with respect to output and income per person between the countries of the world automatically vanish in the long run. This possibility is sufficiently intriguing to have a name of its own:

THE HYPOTHESIS OF ABSOLUTE CONVERGENCE



In the long run GDP per worker (or per capita) converges to one and the same growth path in all countries, so that all countries converge on the same level of income per worker.

7. William J. Baumol, 'Productivity Growth, Convergence and Welfare: What the Long-Run Data Show', *American Economic Review*, 76, 1986, pp. 1072–1085.

FIGURE 2.4 Average annual growth rate of GDP per worker against initial level of GDP per worker, 24 OECD countries¹

Note: The countries behind the country codes can be found in Table A. The straight line is estimated by ordinary least squares (OLS), and it is the uniquely determined line that minimizes the sum of the squared vertical distances between the points in the figure and the line. The estimated line is $y = 11.10 - 0.91x$. Additional information for the statistically proficient: $R^2 = 0.39$, and $t = -3.72$ (standard error of the coefficient is 0.24).

¹ Members in 2003, except Germany, Hungary, Korea, Czech Republic, Poland, and Slovak Republic, which have been excluded due to lack of data.

Source: Penn World Table 6.2.

A main reason why this hypothesis is so fascinating is the implication that poverty should disappear by itself in the long run. It is worth thinking about what this would imply for the relevance of foreign aid policies. It would not make such policies irrelevant altogether. It could be argued that the speed by which poverty disappears is so slow, and the current poverty problems so severe, that foreign aid is currently needed. However, other things being equal, if poverty disappears by itself, it certainly makes foreign aid less necessary than if some countries are caught in poverty traps.

The hypothesis of absolute convergence implies that *countries with relatively low levels of GDP per worker in an initial year will grow relatively fast after that initial year*.⁸ In other words, average growth in GDP per worker from year 0 to year T , say, should be negatively correlated with GDP per worker in year 0. Indexing countries by i , and denoting country i 's GDP per worker in year t by y_t^i , the following equation (where the left-hand side is the approximate average annual growth rate in GDP per worker, β_0 is a constant, and β_1 is a positive parameter) should then be expected to fit well:

$$\frac{\ln y_T^i - \ln y_0^i}{T} \approx \beta_0 - \beta_1 \ln y_0^i \quad (1)$$

Figure 2.4 plots the average annual growth rate of GDP per worker from 1951 to 2003 against the log of GDP per worker in 1951 for all OECD member countries in 2003 for which data were available from the Penn World Table (PWT).

8. It cannot be concluded the other way round that a negative association between initial GDP per worker and subsequent growth is (necessarily) an indication of convergence! See the exercise on Galton's Fallacy at the end of the chapter.

The country points in Fig. 2.4 are nicely located along a decreasing straight line. The line that has been drawn in the figure is a best fit to the points according to a standard statistical method, ordinary least squares (OLS) estimation, which is described in the figure legend and in the statistical appendix to this book. The line has the particular formula:

$$\frac{\ln y_{03}^i - \ln y_{51}^i}{52} \cdot 100 \cong 11.10 - 0.91 \ln y_{51}^i \quad (2)$$

Figures 2.3 and 2.4 seem to support the hypothesis of absolute convergence, but they present data only for some countries. This would not be so much of a problem if those countries were randomly selected and representative. However, they are not. Both figures have, in different ways, selected countries which had relatively high and relatively similar GDPs per worker in 2003. Figure 2.4, for instance, focuses on the club of relatively rich OECD members in 2003. The countries in Fig. 2.4 that had relatively low GDPs per worker around 1950 must have grown relatively fast to be able to join the rich club in 2000. The selection of countries thus has a bias in favour of the hypothesis we are testing. This problem is known as the 'sample selection bias' problem, and was noted in the present context by the economist J. Bradford DeLong in an article in 1988.⁹

To avoid the sample selection bias problem one should try to get a more representative sample. We pick a sample as follows: in order to have good standardized data for a large number of countries, we consider the somewhat shorter period from 1960 to 2003. We include in the sample countries for which data on GDP per worker in 1960 and the average growth rate in GDP per worker between 1960 and 2003 (plus some additional data relevant for later purposes) can be obtained from the PWT. We exclude from the sample countries with less than two million inhabitants in 1960 (to avoid particular small country effects), countries for which oil production accounts for more than 30 per cent of the GDP (to avoid particularities due to very high dependence on oil), and countries for which the data quality is relatively poor according to the PWT itself.¹⁰

This leaves us with a relatively large sample of 65 countries that should be reasonably representative for the countries of the world. Figure 2.5 plots the average growth rate of GDP per worker from 1960 to 2003 against the log of GDP per worker in 1960 for the countries of this sample. The nice negative relationship has disappeared. The slope of the line of best fit has even become positive, but the line fits so poorly that we cannot attach any statistical significance to its slope. Since absolute convergence implies a clear negative relationship, we have to draw the sad conclusion: *the hypothesis of absolute convergence does not hold.*

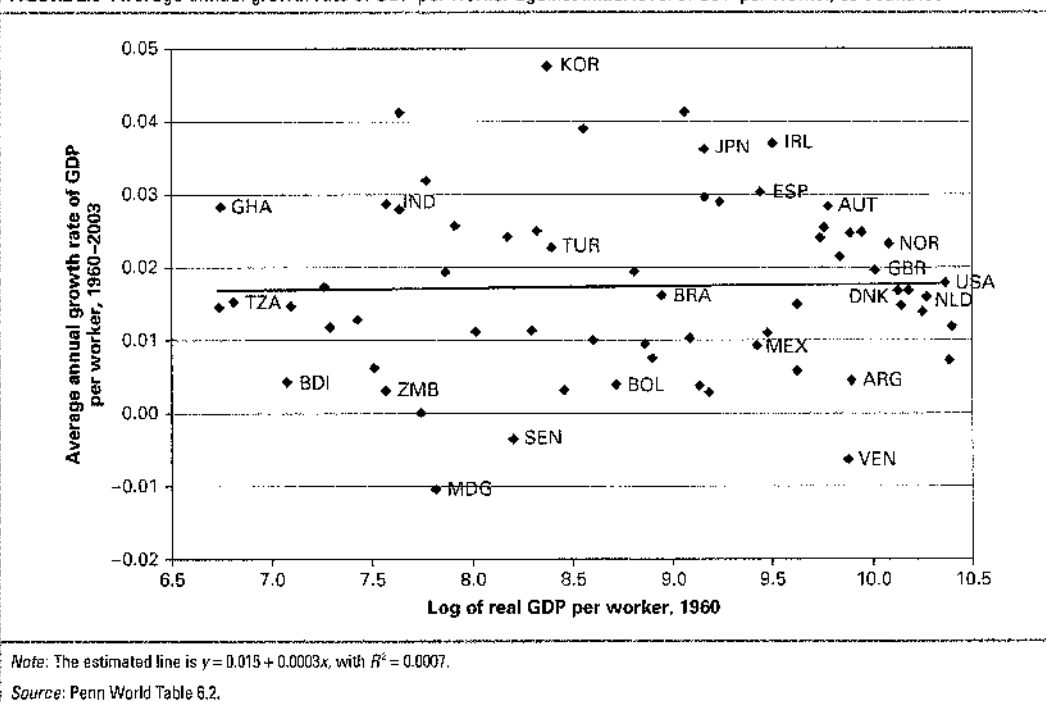
Conditional convergence

Thinking a bit more about it, absolute convergence is too much to hope for. We know that the countries in Fig. 2.5 differ considerably with respect to basic structural characteristics. For instance, some countries have higher rates of saving and investment than others. Savings and investment accumulate as capital, and capital is productive. We should therefore expect countries with higher savings rates to have higher GDP per worker, but then GDP per worker cannot converge to one and the same level for all countries. Similarly, some countries spend a larger fraction of GDP on education (investing in human capital) than others, and education makes labour more productive. Countries with higher investment rates in human capital should therefore be expected to approach higher levels of GDP per worker. A third structural characteristic likely to be important is population growth. Higher population growth means (other things being equal) higher growth of the labour force and therefore, in turn, that a larger

9. J. Bradford DeLong, 'Productivity Growth, Convergence, and Welfare: Comment', *American Economic Review*, 78, 1988, pp. 1138–1154.

10. Each country appearing in the PWT 6.2 is given a grade, A, B, C or D, for data quality. We exclude here countries with grade D.

FIGURE 2.5 Average annual growth rate of GDP per worker against initial level of GDP per worker, 85 countries



number of people will come to share the capital accumulated in the past. Other characteristics being equal, this should pull GDP per capita down, again preventing absolute convergence.

Consider two countries with the same level of GDP per worker in an initial year zero, and suppose that the first country has more favourable structural characteristics than the second. Since the first country will then approach a higher level of GDP per worker than the second, it will also have higher average growth in GDP per worker over a period after year zero. Such reasoning has led to a weaker notion of convergence:

THE HYPOTHESIS OF CONDITIONAL CONVERGENCE



A country's income per worker (or per capita) converges to a **country-specific** long-run growth path which is given by the basic structural characteristics of the country. The further below its own long-run growth path a country starts, the faster it will grow. Income per worker therefore converges to the same level across countries **conditional** on the countries being structurally alike.

Again the convergence hypothesis implies a relationship between the initial level of and the subsequent growth in GDP per worker: *other things (basic structural characteristics) being equal, countries with relatively low levels of GDP per worker in an initial year will grow relatively fast after that initial year.* The crucial addition compared to the absolute convergence hypothesis is the phrase 'other things being equal'. According to the hypothesis of conditional convergence, it is only after controlling appropriately for structural differences that one should find a negative relationship between initial GDP per worker and subsequent growth. The correct equation would not be (1) above, but perhaps an equation like

$$\frac{\ln y_T^i - \ln y_0^i}{T} \cong \beta_0 - \beta_1 \ln y_0^i + \gamma(z^i), \quad (3)$$

where $\mathbf{z}^i$ is a vector of variables capturing country-specific structural characteristics, and γ is a function expressing their influence.

The hypothesis of conditional convergence does not imply that poverty would disappear by itself in the long run. Nevertheless, conditional convergence is also a fascinating possibility. It *does* imply that if a poor country can manage somehow to achieve the same structural characteristics as rich countries, it *will* become as rich in due time. What are the implications of conditional convergence for the relevance of foreign aid? Certainly, conditional convergence gives more room for foreign aid than does absolute convergence, since conditional convergence means that a country may be caught in poverty due to bad structural characteristics. However, the kind of policy suggested is not so much a traditional one of transfers to the poor countries (with a reservation like the one we made in connection with absolute convergence). Rather, the conditional convergence hypothesis points to the importance of supporting poor countries in improving their internal structures. This could, for instance, be a policy of assisting a country in building up sound financial and educational systems.

To work seriously with an equation like (3), for example testing it against data, one must have an idea of how to handle the structural variables. Which economic variables should be included in $\mathbf{z}^i$, and what should the function γ look like? We have argued intuitively above that the rates of investment in physical and human capital and the population growth rate should probably be included, but this does not tell us how they should enter. A good answer will require some growth theory, as presented in the coming chapters. At this stage it will be illustrative to consider a specific version of (3), which is for now postulated, but will be rooted in theory later on. So, assume that (among) the relevant structural characteristics of country i are the rate of investment in physical capital (the GDP share of gross investment in physical capital), s^i , and the growth rate of the labour force, n^i . Assume further that (for reasons that will become clear in Chapters 3 and 5) the appropriate way they enter is:

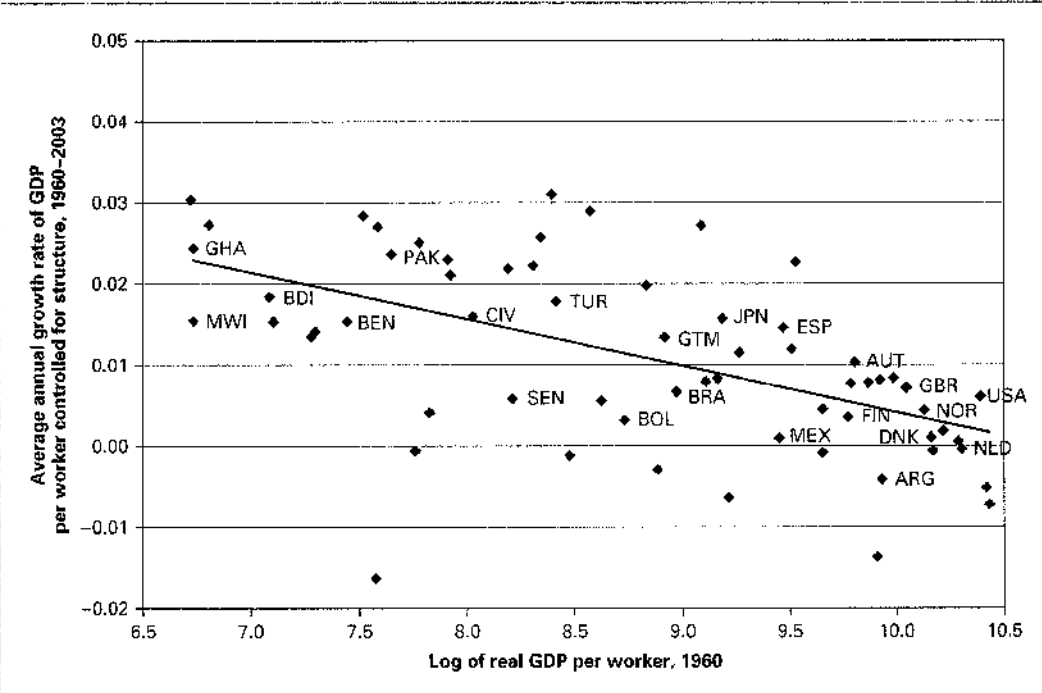
$$\frac{\ln y_T^i - \ln y_0^i}{T} \cong \beta_0 - \beta_1 \ln y_0^i + \beta_2 [\ln s^i - \ln(n^i + 0.075)]. \quad (4)$$

Consider the same countries and years as in Fig. 2.5. Measure s^i and n^i as the country's average gross investment rate and its average labour force growth rate, respectively, over the period 1960 to 2003 (these can also be computed from data in the PWT, and are included in Table A of Book One). Assume that the appropriate value of the parameter β_2 is 0.016 (Chapter 5 will explain this value). Finally, create a diagram like Fig. 2.5, still with $\ln y_{60}^i$ along the horizontal axis, but now measure the growth rate *adjusted for country-specific structural characteristics*, $(\ln y_{03}^i - \ln y_{60}^i)/43 - \beta_2 [\ln s^i - \ln(n^i + 0.075)]$, up along the vertical axis. Figure 2.6 then results.

The points in Fig. 2.6 do seem to be clustered around a negatively sloped straight line. Hence, controlling for *some* structural differences makes the negative relationship between initial GDP per worker and subsequent growth visible again, and we have not even controlled for *all* the structural characteristics that should be important on *a priori* grounds. Education has not been taken into account, but will be later on. This will make the picture even more compatible with conditional convergence. Based on theoretical and empirical work, most economists believe that if one puts the right structural characteristics into $\mathbf{z}^i$, and takes account of these characteristics in the right way (assumes the right γ), then indeed one will end up with a significant and positive estimate of the β_1 in (3). That is, if one controls appropriately for the influence of structural characteristics, growth in GDP per worker is negatively correlated with the initial level of GDP per worker for the countries of the world. This accords with the hypothesis of conditional convergence.

Club convergence

According to the hypothesis of conditional convergence, the long-run growth path of each country is given entirely by the country's structural characteristics. It is independent of the country's initial level of GDP per worker: the starting point has no influence on the long-run

FIGURE 2.6 Average annual growth rate of GDP per worker adjusted for structural characteristics against the initial level of GDP per worker, 65 countries

Source: Penn World Table 6.2.

growth path. Some economists doubt that this is true. They argue that indeed the initial position of a country may have an influence on the level of the growth path that the country is approaching in the long run. In this way history can have a permanent impact – its influence is not washed out in the long run.

What these economists have in mind is that there is a certain threshold value of GDP per worker (or perhaps there are several thresholds, but here we present the idea in its simplest version), which may be country-specific, such that if a country happens to start below that value, it will converge to one growth path, and if it happens to start above, it will converge to another path. The two growth paths may have the same constant growth rate, but they differ with respect to their levels: the first path lies below the second. Figure 2.7 gives an illustration of club convergence.

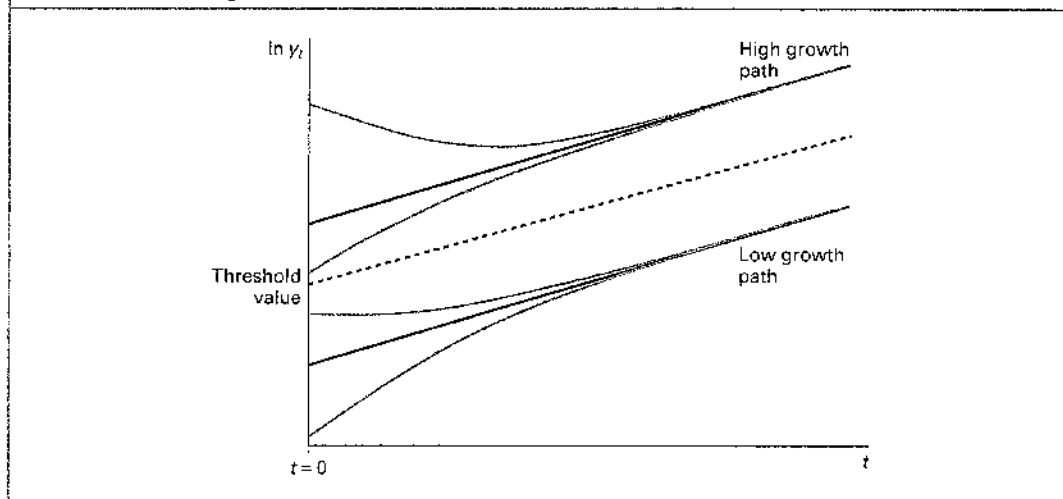
The meaning of the term ‘club’ is that countries that start off on the same side of the threshold value are in the same category. Stating the hypothesis comprehensively, what must be changed from the definition of conditional convergence above is that the long-run growth path depends not only on structural characteristics, but also on the economy’s starting point.

THE HYPOTHESIS OF CLUB CONVERGENCE



A country’s income per worker (or per capita) converges to a long-run growth path that depends on the country’s basic structural characteristics and on whether its initial GDP per capita is above or below a specific threshold value. The further below the relevant growth path a country starts out, the faster it will grow. Income per worker therefore converges to the same level across countries **conditional** on the countries being structurally alike and on the countries starting on the same side of their respective threshold values.

FIGURE 2.7 Club convergence



One can write down sound economic growth models that support the idea of conditional convergence, as well as models supporting the idea of club convergence. Likewise, some empirical analyses are in favour of conditional convergence, and some are in favour of club convergence. At present, the issue of conditional or club convergence is unsettled.

An interesting implication of the idea of club convergence is that it may provide a rationale for traditional foreign aid policy. Giving high transfers to a developing country over some period may bring the country's income per capita above the threshold level, and thereby initiate a growth process eventually leading the country to higher levels of income than would have been reached without the transfers, even after the transfers are no longer given. A foreign aid policy of giving *temporary* transfers to the poor countries can thus have *permanent* beneficial effects, according to the idea of club convergence.

We may sum up our discussion of convergence below in Stylized fact 4. The reader should be warned, however, that the issue of convergence is perhaps a bit more controversial than our use of the word 'fact' suggests:

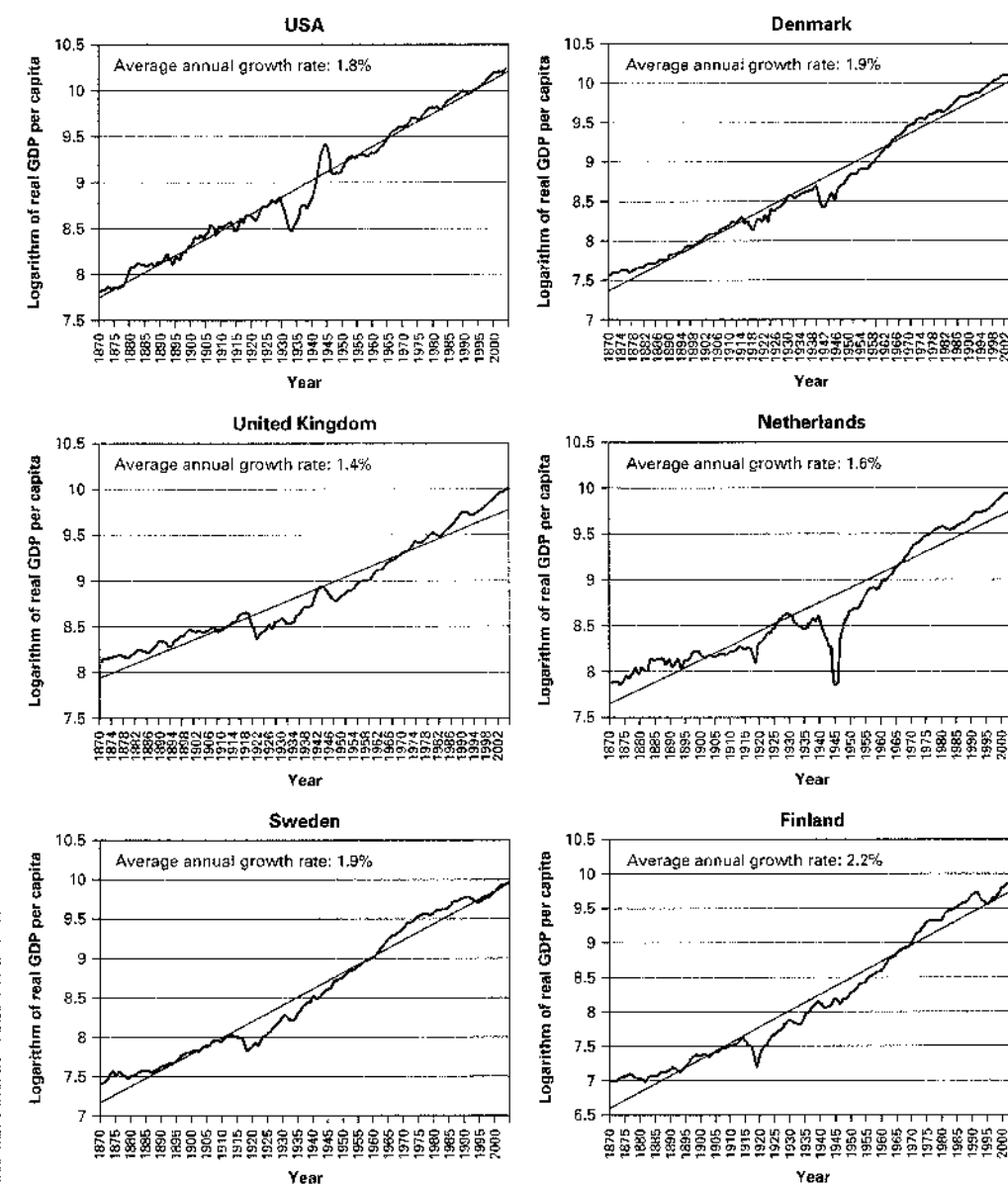
STYLIZED FACT 4

Convergence: If one controls appropriately for structural differences between the countries of the world, a lower initial value of GDP per worker tends to be associated with a higher subsequent growth rate in GDP per worker. This accords with the idea that in the long run, income and GDP per worker converge to a country-specific growth path which is given by the country's basic structural characteristics, and possibly also by its initial position.

2.4 The long-run growth process

The previous section was concerned with the relationship between the growth processes in different countries. Now we will focus on the long-run growth process within a single and steadily growing economy, that is, on growth along the country's own long-run growth path. To separate convergence to the growth path from the long-run growth path itself, we should look at very long series for GDP per person.

FIGURE 2.8 GDP per capita, some Western countries



Source: Angus Maddison, *Monitoring the World Economy 1820–1992*, OECD, 1995 and Penn World Table 6.2.

Steady long-run growth

Figure 2.8 shows the evolution of the log of GDP per capita in a number of Western countries for periods exceeding 100 years, and it also shows the linear trends. Disregarding shorter-run fluctuations and the influences from great wars and depressions, Fig. 2.8 shows for each country a remarkable tendency towards a constant and positive rate of growth in output per capita.

STYLIZED FACT 5

Over periods of more than 130 years, probably up to 200 years, many countries in Western Europe and North America have had relatively constant annual rates of growth in GDP per capita in the range 1.5–2 per cent.

Because of a lack of very long-run data for GDP per worker, Fig. 2.8 shows GDP per capita. However, there is good reason to believe that GDP per worker has also grown at constant (slightly lower) rates in the countries considered, since participation rates have typically increased gradually.

The countries that have experienced relatively constant growth have typically had remarkably constant factor income shares over long periods as well. The ‘law’ of stable income shares is illustrated in Fig. 2.9, which shows the evolution of the income share of labour for six OECD countries, five of which also appear in Fig. 2.8. Not only does labour’s share not show any long-run trend, the relatively constant labour share turns out to be relatively close to 2/3 in all the countries. There are considerable short-run cyclical movements in labour’s share (not so visible from Fig. 2.9), but the *long-run* evolution in the labour income share is fairly precisely described as being constant.

Consider a Western economy that has experienced steady annual growth in GDP per worker, $y_t = Y_t/L_t$, where Y_t and L_t are the GDP and the number of workers in year t , respectively. Labour’s share in year t is $w_t L_t/Y_t$, where w_t is the average real wage per worker. We can rewrite labour’s share as $w_t/(Y_t/L_t) = w_t/y_t$. Hence, if y_t has grown at a relatively constant rate, and labour’s share has stayed relatively constant, the average real wage rate must have been increasing by more or less the same rate as GDP per worker. Steady growth in GDP per worker, and a constant labour income share, implies steady growth in the real wage.

STYLIZED FACT 6

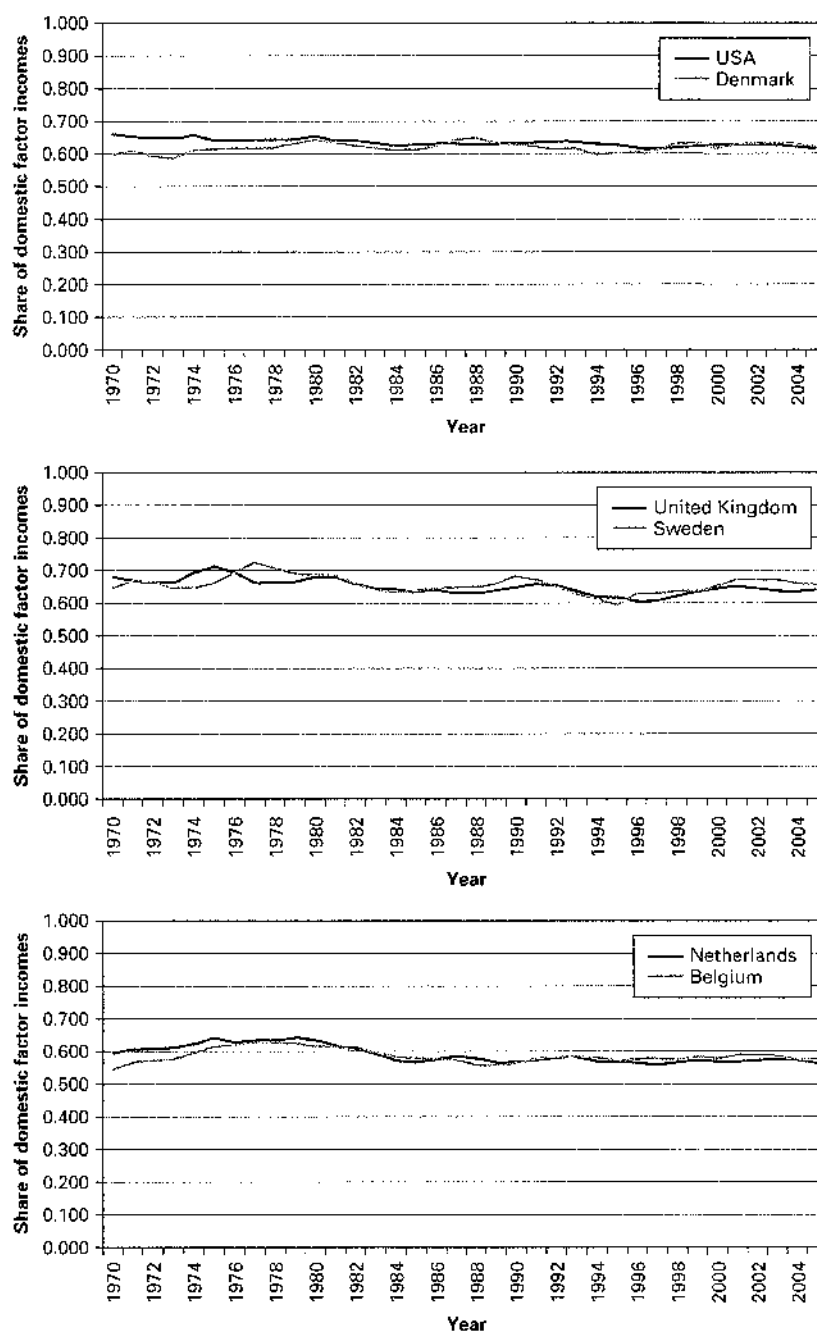
During the long periods of relatively constant growth rates in GDP per worker in the typical Western economy, labour’s share of GDP has stayed relatively constant, and (hence) the average real wage of a worker has grown by approximately the same rate as GDP per worker.

If labour’s share has been relatively constant, so must the share of all other production factors since this latter share is one minus labour’s share. Let us call the other factors ‘capital’, including into this category not only reproducible physical capital, but also, for example, land and other natural resources.

Let the total capital input in year t be denoted by K_t . If we denote the rate of return on capital by r_t , then capital’s share is $r_t K_t/Y_t = r_t/(Y_t/K_t)$. Hence, constancy of capital’s share implies that the real rate of return on capital, r_t , and the output–capital ratio, Y_t/K_t , must have been changing by the same rates.

Over long periods there should be no systematic differences between the trends in the real rates of return on different types of assets. If the rate of return on capital (as defined here) over a long period increased by one half of a per cent per year, say, while over the same period the real rate of interest on bonds stayed constant, investment in real capital would soon become much more advantageous than buying bonds. This would direct portfolios away from bonds and towards real capital, which would tend to equalize the rates of returns between bonds and capital.

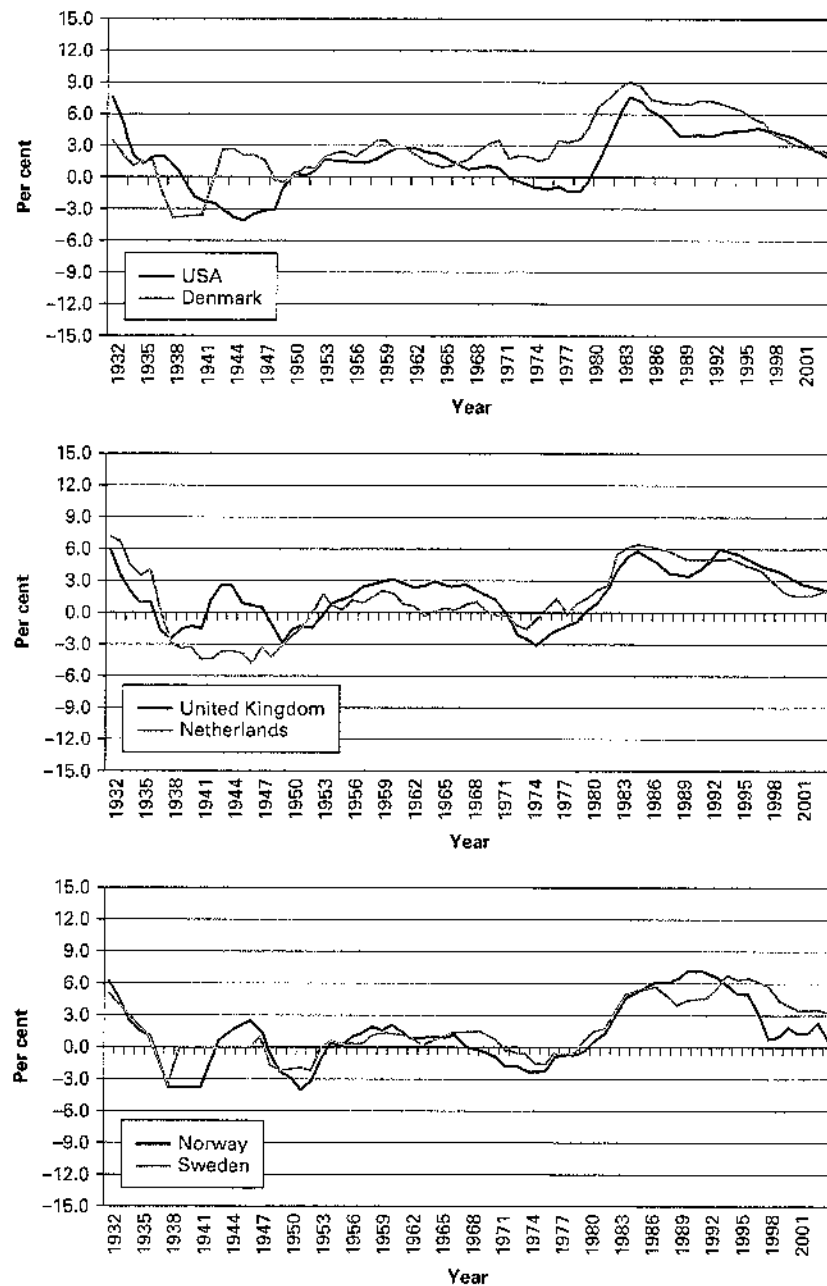
In the long run the trend of the rate of return on capital must, therefore, be anchored by the trend of real interest rates on bonds. Figure 2.10 reports on the long-run behaviour of real interest rates over long periods in some Western countries. The figure shows that real

FIGURE 2.9 Labour's share of domestic factor incomes

Note: Labour income is compensation to employees. Domestic factor income is compensation to employees plus gross operating surplus and gross mixed income.

Source: OECD, Annual National Accounts.

FIGURE 2.10 Real interest rates, Western countries



Note: The formula used for computing annual real interest rates is $(1 + r)(1 + p) = 1 + i$, where r is the real interest rate, i is the nominal interest rate, and p is the inflation rate measured as the relative change in average consumer prices from the current year to the next. The nominal interest rate used is the average annual yield on long-term government bonds. The figure shows five year moving averages of real interest rates computed this way.

Sources: Up to 1990–94, interest rates are from S. Homer and R. Sylta, *A History of Interest Rates*, Rutgers University Press, 1991, and consumer price indices are from B.R. Mitchell, *International Historical Statistics, 1750–1993*, Macmillan, 1998. For recent years interest rates and consumer price indices are from Source OECD.

interest rates (as computed in the figure) fluctuate, but there is no tendency for real interest rates to be systematically increasing or decreasing over long periods: they have no long-run trend, upwards or downwards. For purposes of long-run analysis, real interest rates, and hence the real rate of return on capital, may therefore be treated as if they were constant.

If capital's share, $r_t/(Y_t/K_t)$, and the rate of return on capital, r_t , have both been relatively constant, then the output–capital ratio, Y_t/K_t , and the capital–output ratio, K_t/Y_t , must also have been constant. We can rewrite the capital–output ratio as $K_t/Y_t = (K_t/L_t)/(Y_t/L_t) = k_t/y_t$, where we have denoted the capital–labour ratio, or capital intensity, K_t/L_t , by k_t . Constancy of K_t/Y_t implies that the capital intensity grows at the same rate as GDP per worker.

STYLIZED FACT 7

During the long periods of relatively constant growth in GDP per worker in the typical Western economy, capital's share and the rate of return on capital have shown no trend, (therefore) the capital–output ratio has been relatively constant, and the capital intensity has grown by approximately the same rate as GDP per worker.

Balanced growth

The empirical regularities in our Stylized facts 5–7 are much inspired by a famous lecture given by the British economist Nicholas Kaldor, who pioneered the approach of setting up the stylized facts and constructing theories to explain them.¹¹ All three facts can be expressed comprehensively by three fundamental constancies: the growth rate of GDP per worker is relatively constant, the functional income distribution between labour and 'capital' is relatively constant, and the rate of return on 'capital' is relatively constant. All the other features in our list of facts follow from these constancies (convince yourself of this).

The stylized facts have given rise to an idealized picture of the long-run growth process called 'balanced growth'. Consider an economy that fulfils the three constancies with an annual growth rate, g , in GDP per worker, a constant annual growth rate, n , in the number of workers, and for which total annual consumption is a constant fraction, $1 - s$, of total annual GDP (a realistic long-run feature for a typical western economy).

Since $Y_t = y_t L_t$, one has $\ln Y_t - \ln Y_{t-1} = (\ln y_t - \ln y_{t-1}) + (\ln L_t - \ln L_{t-1}) \approx g + n$, so GDP grows at a constant rate equal to the sum of the growth rate of GDP per worker and the population growth rate. Hence total consumption, $(1 - s)Y_t$, and total investment, sY_t , must also be growing at the rate, $g + n$. Finally, since $K_t = k_t L_t$, and k_t is growing at the same rate as y_t , total capital is growing at the rate, $g + n$.

BALANCED GROWTH¹²



The growth process follows a balanced growth path if GDP per worker, consumption per worker, the real wage rate, and the capital intensity all grow at one and the same constant rate, g , the labour force (population) grows at constant rate, n , GDP, consumption, and capital grow at the common rate, $g + n$, the capital–output ratio is constant, and the rate of return on capital is constant.

11. You can find this lecture by Nicholas Kaldor, 'Capital Accumulation and Economic Growth', in *The Theory of Capital*, F.A. Lutz and D.C. Hague (eds), New York, St Martins, 1961.

12. Our definition of balanced growth is a relatively strict one. Often balanced growth is just meant to describe a situation where GDP, consumption, capital and population all grow at constant rates, which is implied by, but does not itself imply, our definition. We find the definition given here useful for how we are going to use the concept of balanced growth: as a comprehensive expression of the empirical regularities that we want our growth theories to explain.

Along a balanced growth path the capital–output ratio is constant. In fact, the constancy of the capital–output ratio is a main motivating factor behind the definition of balanced growth. We have argued indirectly for this constancy: it follows from constant income shares and the absence of a trend in the rate of return on capital, where we argued for the latter by pointing to the long-run behaviour of real interest rates. However, it should be mentioned that there is some controversy about whether long-run capital–output ratios are really constant. Direct estimates of the long-run evolution of capital–output ratios, most notably some by the British economist and economic historian Angus Maddison, suggest that the capital–output ratio may be close to constant in the US in the long run, but in several other countries the ratio seems to have been gradually increasing over the past 100 years.¹³

When we turn to growth models in the subsequent chapters, balanced growth will be of theoretical importance. If a growth model predicts that the economy will converge to or move along a specific long-run growth path, then this path should be a balanced growth path, because of the empirical plausibility of balanced growth. Long-run accordance with balanced growth will thus be used as an ‘empirical consistency check’ of growth models.

2.5 Summary



1. A country’s GDP per worker may be used as a proxy for the average standard of living and the average productivity of labour in the country. In some countries the population participates much less in the formal market economy than in other countries. Focusing on GDP per member of the official labour force rather than on GDP per capita is a rough way of adjusting for this.
2. The empirical evidence on levels of GDP per worker reveals enormous income differences across countries. Combining data on GDP per worker with data on population size, we found that the relative cross-country income differences have declined over the past 30 years, but there was little reduction in international inequality at the very bottom of the world income ladder. The fact that the world income distribution has improved *in relative terms* means that *on average* poor countries have enjoyed percentage increases in income per worker at or above the percentage increase in income per worker in the world in general. *In absolute terms* poverty has therefore become less severe, but at the same time the *absolute* international income differences between the richest and the poorest countries have increased.
3. The evidence shows that growth rates vary substantially across countries. By the process of growing or declining quickly, a country can move from being relatively poor to being relatively rich, or from being relatively rich to being relatively poor. Moreover, growth can break in a country, turning from a high rate to a low one, or vice versa. These facts indicate that the fight against poverty is not hopeless. On the other hand, a relatively prosperous country will not automatically remain so regardless of the policies it follows. These observations point to the importance of structural policies to promote or maintain economic growth.
4. The evidence seems to support the hypothesis that the world’s countries converge in a conditional sense: if one controls appropriately for structural differences across countries, a lower initial value of GDP per worker tends to be associated with a higher subsequent growth rate in GDP per worker. This accords with the idea that in the long run, income and GDP per worker converge to a country-specific growth path which is given by the country’s basic structural characteristics (and possibly also by its initial position).

13. Angus Maddison, *Dynamic Forces in Capitalist Development: A Long Run Comparative View*, Oxford University Press, 1991.

5. Over periods of more than 130 years, and probably up to 200 years, many countries in Western Europe and North America have had relatively constant annual rates of growth in GDP per capita in the range 1.5–2 per cent. During these long periods, labour's share of GDP has stayed relatively constant. Hence the average real wage of a worker has grown by approximately the same rate as GDP per worker. Furthermore, capital's share and the rate of return on capital have shown no trend, so the capital–output ratio has been relatively constant, that is, the stock of capital per worker has grown at roughly the same rate as GDP per worker.
6. The facts just mentioned have given rise to an idealized picture of the growth process called 'balanced growth'. In this book balanced growth will describe a situation where GDP per worker, consumption per worker, the (average) real wage rate, and the capital–labour ratio all grow at one and the same constant rate, and the rate of return on capital is constant. We will use long-run accordance with balanced growth as an empirical consistency check for the growth models to be presented in coming chapters.

2.6 Exercises

Exercise 1. GDP per capita versus GDP per worker

Give some arguments why GDP per worker, rather than GDP per capita, should be used for a comparison of standards of living between countries.

Exercise 2. The income distribution for the rich part of the world

Table 2.4 shows GDP per worker (in 2000 US dollars) and population size for 1951 and 2003 for the rich part of the world (defined here as the OECD countries for which data were available).

Construct Lorenz curves for the rich world for 1951 and 2003, respectively, assuming that everybody in the population earned the GDP per worker of the relevant year, and draw the two curves in one figure. Compute the Gini coefficient for each year. Comment on how inequality, or equality, among the rich has developed. Compare this to how the income distribution of the world has developed. Relate this to what you know about convergence from Section 2.3.

Exercise 3. Time to double

Assume that in a specific country GDP per capita, or per worker, grows at the rate g each year over many years, where g is written as a percentage, e.g. 2 per cent. Show that GDP per capita, or worker, will then approximately double every $70/g$ years. Use this to set up a table showing for each of the 15 richest countries in the world in 2003, the country's average annual growth rate in GDP per worker from 1960 to 2003 (you can take these from Table 2.2), and the number of years it takes for the country's GDP per worker to double, if growth continues at the same rate as it averaged from 1960 to 2003.

Exercise 4. Growth against ultimo level

In Section 2.3 there were figures showing across countries the log of GDP per worker in an initial year along the horizontal axis, and average annual growth in GDP per worker in a subsequent relatively long period along the vertical axis. For the OECD countries there was a clear negative relationship (see Fig. 2.4). Construct an 'opposite' figure for the same countries as in Fig. 2.4, where along the horizontal axis you have the log of GDP per worker, not in an initial year, but in a final year, 2003, and along the vertical axis you have the average annual growth rate in GDP per worker from 1960 to 2003. Comment on your findings. Data for this exercise can be taken from Table A at the end of Book One.

TABLE 2.4. Population size and GDP per worker, 1951 and 2003: OECD countries¹

	1951			2003	
	Population (1,000)	GDP per worker (2000 dollars)		Population (1,000)	GDP per worker (2000 dollars)
Turkey	22,048	3,248	Turkey	71,252	11,812
Japan	85,118	5,496	Mexico	104,337	18,628
Portugal	8,393	6,312	Portugal	10,386	34,076
Greece	7,647	7,049	Greece	11,075	35,945
Spain	28,252	8,733	New Zealand	3,946	44,347
Mexico	28,438	8,996	Japan	127,736	45,030
Austria	8,930	10,435	Iceland	289	46,240
Italy	47,407	10,441	Spain	42,144	46,619
Ireland	2,966	10,629	Finland	5,220	48,016
Finland	4,050	11,358	Sweden	8,970	48,128
France	42,034	13,620	Denmark	5,397	51,499
Iceland	145	14,212	Canada	31,636	51,796
Belgium	8,673	16,213	United Kingdom	59,282	51,924
Denmark	4,310	17,355	Italy	57,961	52,097
Norway	3,299	17,541	Switzerland	7,226	54,256
United Kingdom	49,862	18,243	Australia	19,728	54,600
Netherlands	10,230	19,057	Netherlands	16,148	56,790
Sweden	7,076	19,736	France	60,008	56,909
Luxembourg	297	22,090	Austria	8,150	59,789
Australia	8,414	22,108	Belgium	10,379	61,541
Canada	14,133	24,118	Norway	4,575	65,699
New Zealand	1,948	24,834	Ireland	4,007	65,825
Switzerland	4,738	25,489	United States	282,617	67,865
United States	160,016	27,426	Luxembourg	453	117,759

¹Members in 2003 except the Czech Republic, the Slovak Republic, Hungary, Poland, Korea & Germany.

Source: Penn World Table 6.2.

Exercise 5. Level convergence among the rich?

Commenting on Fig. 2.3 in Section 2.3 we said that it showed a tendency towards convergence to a common growth path with constant growth for the countries involved. Perhaps we were too hasty. The following figure will make level differences more visible if they are there. For the same countries and years as in Fig. 2.3, compute GDP per worker relative to GDP per worker in the USA over the period and illustrate in a figure with *relative* GDP per capita against years. The US will appear as a horizontal line. Comment on the issue if all countries other than the US seem to be approaching the US horizontal line. To find the data for this exercise, you will have to go to the PWT 6.2 yourself (<http://pwt.econ.upenn.edu/>).

Exercise 6. Growth and trade

This chapter has presented the stylized facts of growth that are most important for the theoretical developments in subsequent chapters. There is at least one more important fact, which relates growth to growth in trade. To draw the following figure you should go to the PWT 6.2 yourself (<http://pwt.econ.upenn.edu/>). First make sure that you understand what

the variable 'openness' measures. For this you will need to go to the explanatory documents of the PWT. Across as many countries as possible, plot the average annual growth rate of GDP (not per worker) from 1960 to 2003, against the average annual growth rate of 'trade volume', where the latter is defined as the total value of exports and imports. Do an OLS estimation of the line of best fit. What has grown most, world GDP or world trade? Can it be inferred from the figure whether growth in trade causes growth in GDP, or growth in GDP causes growth in trade?

Exercise 7. Galton's Fallacy¹⁴

It is an implication of convergence that a lower initial GDP per worker should, other things being equal, imply a higher subsequent growth in GDP per worker. This was used in the chapter to test convergence: the lack of a negative relationship between initial GDP per worker and subsequent growth would imply a rejection of the hypothesis. Is it also true the other way around, that a clear negative relationship between initial GDP per worker and subsequent growth is necessarily an indication of convergence? The answer is no, as this exercise should teach you.¹⁵

To each of 26 imaginary countries attribute a GDP per worker in 1960 (measured in US 2000 dollars, say) as follows: 10,000 to Country 1, 12,000 to Country 2, 14,000 to Country 3, etc., up to 60,000 to Country 26. Now determine each country's GDP per worker in the year 2003 by *picking randomly* (without replacement) from the same set of values for GDP per worker, {10,000, 12,000, . . . , 60,000}, one value for each country (you can do this on the computer or by drawing values written on pieces of paper from a box). For each country you will now have an initial 1960 value for GDP per worker and an eventual year 2003 value. Plot across countries the average annual growth rate in GDP per worker against the initial level of GDP per worker. Why is it that there must be a negative relationship? Can we infer anything about convergence from this negative relationship?

14. The term 'Galton's Fallacy' comes from the English statistician Francis Galton who once, late in the nineteenth century, made a similar fallacy in connection with heights of sons and fathers. Galton noted a tendency that relatively tall fathers got sons shorter than themselves, and vice versa. He concluded from this, a tendency to 'convergence' towards a general intermediate height of everybody.

15. The exercise is based on Danny Quah, 'Galton's Fallacy and Tests of the Convergence Hypothesis', *Scandinavian Journal of Economics*, 95, 1993, pp. 427–443.

PART

1

Basic Theory and Empirics about Prosperity and Growth

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